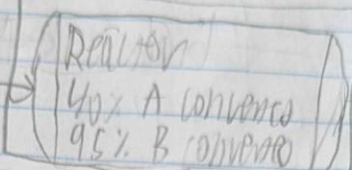


HW #6

1. $\text{CH}_3\text{OH} = \text{A}$, $\text{O}_2 = \text{B}$, $\text{HCHO} = \text{C}$, $\text{CO} = \text{D}$, $\text{CO}_2 = \text{E}$, $\text{H}_2\text{O} = \text{F}$, $\text{HCOOH} = \text{G}$

21.05 $\frac{\text{kmol}}{\text{hr}}$ B

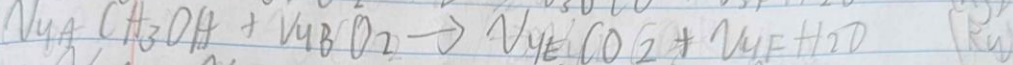
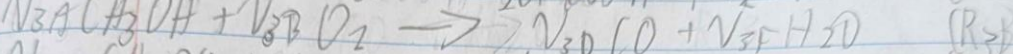
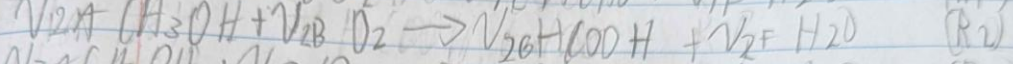
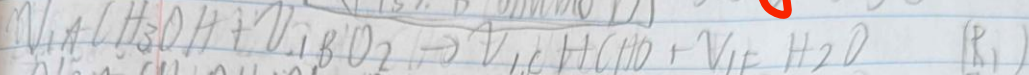
(A, B)



SYSTEM

A, B, C, D, E, F, G

(D)



$$C_1: V_{1A} = V_{1C}$$

$$C_2: V_{2A} = V_{2G}$$

$$H_1: 4V_{1A} = 2V_{1C} + 2V_{1F}$$

$$H_2: 4V_{2A} = 2V_{2G} + 2V_{2F}$$

$$O_1: V_{1A} + 2V_{1B} = V_{1C} + V_{1F}$$

$$O_2: V_{2A} + 2V_{2B} = 2V_{2G} + V_{2F}$$

$$\text{let } V_{1A} = 1$$

$$\text{let } V_{2A} = 1$$

$$V_{1C} = 1$$

$$V_{2G} = 1$$

$$4(1) = 2(1) + 2V_{1F}$$

$$4(1) = 2(1) + 2V_{2F}$$

$$2 = 2V_{1F}$$

$$1 = V_{2F}$$

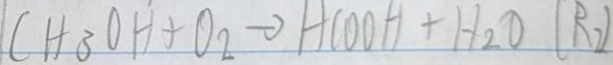
$$1 = V_{1F}$$

$$1 + 2V_{2B} = 2(1) + 1$$

$$1 + 2V_{1B} = 1 + 1$$

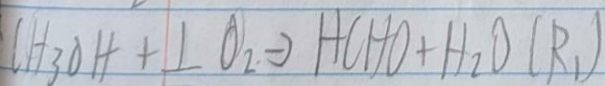
$$V_{2B} = 1$$

$$2V_{1B} = 1$$



$$V_{1B} = \frac{1}{2}$$

$$\frac{1}{2}$$



$$\frac{1}{2}$$

$$C_3: V_{3A} = V_{3D}$$

$$H_3: 4V_{3A} = 2V_{3F}$$

$$O_3: V_{3A} + 2V_{3B} = V_{3D} + V_{3F}$$

$$1 \text{ pt } V_{3A} = 1$$

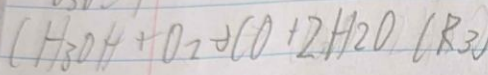
$$V_{3D} = 1$$

$$4(1) = 2V_{3F}$$

$$V_{3F} = 2$$

$$1 + 2V_{3B} = 1 + 2$$

$$V_{3B} = 1$$



$$C_4: V_{4A} = V_{4E}$$

$$H_4: 4V_{4A} = 2V_{4F}$$

$$O_4: V_{4A} + 2V_{4B} = 2V_{4E} + V_{4F}$$

$$1 \text{ pt } V_{4A} = 1$$

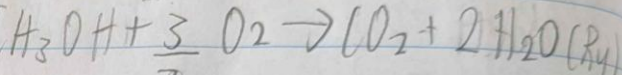
$$V_{4E} = 1$$

$$4(1) = 2V_{4F}$$

$$V_{4F} = 2$$

$$1 + 2V_{4B} = 2(1) + 2$$

$$V_{4B} = \frac{3}{2}$$



$$DOF = (4q + 4b) - (5d + 5e + 5c + 5a) = (9 + 4) - (2 + 0 + 2 + 7) = 13 - 11$$

$\Rightarrow 2$ unspecified

$$\frac{100 \text{ kg mol } A_f}{h} \left(\frac{(100 - 40) \text{ kg mol } A_p}{100 \text{ kg mol } A_f} \right) = 60 \text{ kg mol } A_p$$

$$\frac{21.05 \text{ kg mol } B_f}{h} \left(\frac{(100 - 95) \text{ kg mol } B_p}{100 \text{ kg mol } B_f} \right) = 1.0525 \text{ kg mol } B_p$$

species mole balances:

$$A: \dot{n}_{A,out} = \dot{n}_{A,in} - \dot{S}_1 - \dot{S}_2 - \dot{S}_3 - \dot{S}_4$$

$$B: \dot{n}_{B,out} = \dot{n}_{B,in} - \frac{1}{2} \dot{S}_1 - \dot{S}_2 - \dot{S}_3 - \frac{3}{2} \dot{S}_4$$

$$C: \dot{n}_{C,out} = \dot{S}_1$$

$$D: \dot{n}_{D,out} = \dot{S}_3$$

$$E: \dot{n}_{E,out} = \dot{S}_4$$

$$F: \dot{n}_{E,out} = \dot{\xi}_1 + \dot{\xi}_2 + 2\dot{\xi}_3 + 2\dot{\xi}_4$$

$$G: \dot{n}_{G,out} = \dot{\xi}_2$$

$$Eq(A) - Eq(B): \dot{n}_{A,out} - \dot{n}_{B,out} = \dot{n}_{A,in} - \dot{n}_{B,in} - \frac{1}{2} \dot{\xi}_1$$

$$+ \frac{1}{2} \dot{\xi}_4$$

$$\text{Let } \dot{\xi}_4 = 0$$

$$60 - 1.0525 = 100 - 21.05 - \frac{1}{2} \dot{\xi}_1 + \frac{1}{2} (0)$$

$$R_4: 58.9475 = 78.95 - \frac{1}{2} \dot{\xi}_1$$

$$\frac{1}{2} \dot{\xi}_1 = 20$$

$$\dot{\xi}_1 = 40 \text{ eq/hr}$$

$$60 = 100 - 40 - \dot{\xi}_2 - \dot{\xi}_3$$

$$0 = \dot{\xi}_2 + \dot{\xi}_3$$

since no products are entering the reactor, the reactions cannot be going backwards, so the extents of reaction cannot be negative

$$\dot{\xi}_2 = \dot{\xi}_3 = 0$$

$$\dot{n}_{C,out} = 40 \text{ kmol/h}$$

$$\dot{n}_{E,out} = 40 \text{ kmol/hr}$$

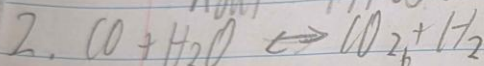
$$n_{out} = n_{A,out} + n_{B,out} + n_{C,out} + n_{H_2O,out} = 60 + 1.0525 + 40 + 40 = 141.0525 \text{ kmol/h}$$

$$x_{A,out} = \frac{n_{A,out}}{n_{out}} = \frac{60}{141.0525} = 0.425$$

$$x_{B,out} = \frac{n_{B,out}}{n_{out}} = \frac{1.0525}{141.0525} = 0.007$$

$$x_{C,out} = \frac{n_{C,out}}{n_{out}} = \frac{40}{141.0525} = 0.284$$

$$x_{H_2O,out} = \frac{n_{H_2O,out}}{n_{out}} = \frac{40}{141.0525} = 0.284$$

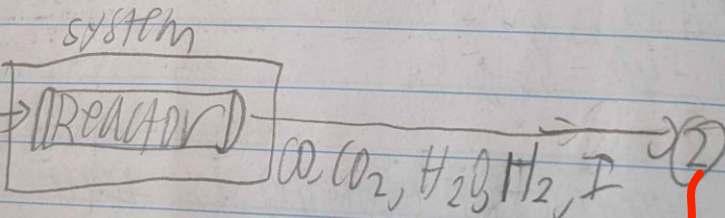


$$K_{CO_2, H_2} = K_e(T) = a e^{\frac{b}{T}}, a = 0.0247, b = 4020 \text{ K}$$

$$T = 1123 \text{ K}$$

$$(a) \quad n_1 = 1 \text{ mol}$$

$$\begin{aligned} & 20 \text{ mol\% CO,} \\ & 10 \text{ mol\% CO}_2, \\ & 40 \text{ mol\% H}_2\text{O,} \\ & I \end{aligned}$$



$$\text{DOF} = (4a + 4b) - (5a + 5b + 5c + 5d) = (4 + 1) - (1 + 3 + 1 + 5) = 0 \text{ well-posed}$$

HW #6 c.

2.1. (b) It's a batch reactor, there are 2 moles on each side of the reaction, and the moles do not change, so the total moles of gas in the reactor at equilibrium are the total moles of feed, which is 1 mol.

species mole balances:

$$\text{CO: } n_{\text{CO,out}} = n_{\text{CO,in}} - \xi$$

$$\text{H}_2\text{O: } n_{\text{H}_2\text{O,out}} = n_{\text{H}_2\text{O,in}} - \xi$$

$$\text{CO}_2: n_{\text{CO}_2,\text{out}} = n_{\text{CO}_2,\text{in}} + \xi$$

$$\text{H}_2: n_{\text{H}_2,\text{out}} = \xi$$

$$\text{T: } n_{\text{T,out}} = n_{\text{T,in}}$$

$$n_{\text{CO,in}} = 0.2(1) = 0.2 \text{ mol}, \quad n_{\text{H}_2\text{O,in}} = 0.4(1) = 0.4 \text{ mol}$$

$$n_{\text{CO}_2,\text{in}} = 0.1(1) = 0.1 \text{ mol}, \quad n_{\text{T,in}} = (1 - 0.2 - 0.4 - 0.1)(1) = 0.3 \text{ mol}$$

$$0.0247 e^{\frac{1173}{T}} = \frac{X_{\text{CO}_2} X_{\text{H}_2}}{X_{\text{CO}} X_{\text{H}_2\text{O}}}$$

$n_{\text{out}} = n_{\text{in}} = 1 \text{ mol}$ (moles are conserved in this case, as explained above)

$$X_{\text{CO}} = \frac{n_{\text{CO,out}}}{n_{\text{out}}} = \frac{n_{\text{CO,in}} - \xi}{n_{\text{out}}}, \quad X_{\text{H}_2} = \frac{\xi}{n_{\text{out}}}, \quad X_{\text{CO}_2} = \frac{n_{\text{CO}_2,\text{in}} + \xi}{n_{\text{out}}}$$

$$X_{\text{H}_2\text{O}} = \frac{n_{\text{H}_2\text{O,in}} - \xi}{n_{\text{out}}}$$

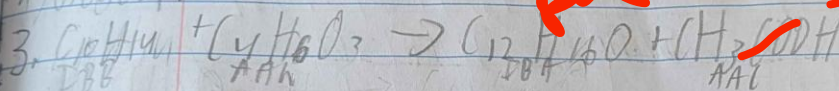
$$0.0247 = \frac{\left(\frac{0.1 + \xi}{1}\right) \left(\frac{\xi}{1}\right)}{\left(\frac{0.2 - \xi}{1}\right) \left(\frac{0.4 - \xi}{1}\right)} = \frac{(0.1 + \xi)(\xi)}{(0.2 - \xi)(0.4 - \xi)}$$

$\xi = 5.64006 \text{ mol}, 0.11002 \text{ mol}$
 There is no H_2 entering the system, so the reaction will start moving forwards until it reaches equilibrium. Thus, the highest extent is nonphysical.

$$n_{\text{H}_2, \text{out}} = 0.1102 \text{ mol}$$

$$x_{\text{H}_2, \text{out}} = \frac{n_{\text{H}_2, \text{out}}}{n_{\text{out}}} = \frac{0.1102}{1} = 0.1102 \text{ mol H}_2 / \text{mol}$$

(c) Shortly after startup, the system will likely not have reached equilibrium yet, so the mole fraction of hydrogen will not be the equilibrium mole fraction of hydrogen.



134 g IBB, 134 g AAN System 27 g IBB, 52 g AAN, 121 g IBA
 (In) (REACTOR) 58 g AAN (Out)

$$n_{\text{IBB}, \text{in}} = 134 \text{ g IBB} \left(\frac{1 \text{ mol IBB}}{134.22 \text{ g IBB}} \right) = 0.998 \text{ mol IBB}$$

$$n_{\text{AAN}, \text{in}} = 134 \text{ g AAN} \left(\frac{1 \text{ mol AAN}}{102.09 \text{ g AAN}} \right) = 1.313 \text{ mol AAN}$$

$$n_{\text{IBB}, \text{out}} = 27 \text{ g IBB} \left(\frac{1 \text{ mol IBB}}{134.22 \text{ g IBB}} \right) = 0.201 \text{ mol IBB}$$

$$n_{\text{AAN}, \text{out}} = 52 \text{ g AAN} \left(\frac{1 \text{ mol AAN}}{102.09 \text{ g AAN}} \right) = 0.509 \text{ mol AAN}$$

$$n_{\text{IBBA}} = 121.4 \text{ g IBBA} \left(\frac{1 \text{ mol IBBA}}{176.26 \text{ g IBBA}} \right) = 0.686 \text{ mol IBBA}$$

$$n_{\text{AAc}} = 68 \text{ g AAc} \left(\frac{1 \text{ mol AAc}}{60.05 \text{ g AAc}} \right) = 1.132 \text{ mol AAc}$$

$$\text{IBB: } 0.201 \text{ mol} = 0.148 \text{ mol} - \text{S}$$

$$\text{AAc: } 0.504 \text{ mol} = 1.313 \text{ mol} - \text{S}$$

$$\text{IBBA: } 0.686 \text{ mol} = \text{S}$$

$$\text{AAc: } 1.132 \text{ mol} = \text{S}$$

0.686 ≠ 1.132 The chemist's analysis is wrong.

4. atomic balance: (EA)

20% excess air

FURNACE

2 mol % C_3H_8 , 79 mol % N_2
 4 mol % O_2
 1 mol % CO
 $\text{N}_2, \text{O}_2, \text{H}_2\text{O}, \text{SO}_2, \text{CO}, \text{H}_2$

(CW)

2 mol % C_3H_8

14 mol % C_4H_{10}

36 mol % C_4H_{10} , 20 mol % C_2H_6

13 mol % N_2 , 4 mol % O_2

8 mol % O_2 , 3 mol % H_2S

$$\text{DOF} = (4 + 4) - (5 + 5 + 5) = (16 + 0) - (10 + 8 + 2 + 5) = 16 - 15 = 1 \text{ underspecified}$$

Basis: $n_{\text{CW}} = 100 \text{ mol}$

$$\text{C: } 100 [(0.02)(3) + (0.14)(4) + (0.36)(1) + (0.2)(2) + (0.08)(1)] = n_{\text{EF}} [(x_{\text{CO,EF}})(1) + (x_{\text{CO}_2\text{,EF}})(1)]$$

$$100 (0.06 + 0.56 + 0.36 + 0.4 + 0.08) = n_{\text{EF}} (x_{\text{CO,EF}} + 4x_{\text{CO}_2\text{,EF}})$$

$$\begin{aligned}
 146 &= 5 n_{EF} \chi_{O_2,EF} \\
 224.2 &= n_{EF} \chi_{O_2,EF} \\
 H: 100 (0.02)(6) + (0.14)(6) + (0.36)(4) + (0.2)(6) + (0.03)(2) \\
 &= n_{EF} (\chi_{H_2O,EF})(2) \\
 100 (0.12 + 0.84 + 1.44 + 1.2 + 0.06) &= 2 n_{EF} \chi_{H_2O,EF} \\
 366 &= 2 n_{EF} \chi_{H_2O,EF} \\
 183 &= n_{EF} \chi_{H_2O,EF} \\
 O: 100 (0.02)(1) + (0.14)(2) + (0.20)(2) + (0.08)(1) \\
 + n_{EA} (0.21)(2) &= n_{EF} (\chi_{O_2,EF})(2) + (\chi_{H_2O,EF})(1) \\
 &+ (\chi_{SO_2,EF})(2) + (\chi_{CO,EF})(1) + (\chi_{CO_2,EF})(2) \\
 100 (0.02 + 0.42 + 0.08 + 0.08) + 0.42 n_{EA} &= n_{EF} (2 \chi_{O_2,EF} \\
 + \chi_{H_2O,EF} + 2 \chi_{SO_2,EF} + \chi_{CO,EF} + 8 \chi_{CO_2,EF}) \\
 60 + 0.42 n_{EA} &= n_{EF} (2 \chi_{O_2,EF} + \chi_{H_2O,EF} + 2 \chi_{SO_2,EF} \\
 + 9 \chi_{CO_2,EF}) \\
 N: 100 (0.13)(2) + n_{EA} (0.74)(2) &= n_{EF} (\chi_{N_2,EF})(2) \\
 26 + 1.68 n_{EA} + 2 n_{EF} \chi_{N_2,EF} \\
 S: 100 (0.03)(1) &= n_{EF} (\chi_{SO_2,EF})(1) \\
 3 &= n_{EF} \chi_{SO_2,EF}
 \end{aligned}$$

Final comps.
 — 10 Q's
 1575